BCS-083	IT in Forensics Science	L T P 3 0 0	3 Credits

Pre-requisites: Students should know about the symmetric and non symmetric technique.

Course Objectives:

CO1	To understand the nature, meaning and significance of forensic science and its relation to crime and criminal.
CO2	To Understand the knowledge of Biometrics
CO3	To learn the Principles of Steganography
CO4	To understand Watermarking and Copyright Protection
CO5	To understand the concept of Data recovery

Detailed Syllabus

MODULE-I

Overview of Biometrics: Biometric Identification, Biometric Verification, Biometric Enrollment, Biometric System Security. Authentication and Biometrics: Secure Authentication Protocols, Access Control Security Services, Matching Biometric Samples, Verification by humans. Common biometrics: Finger Print Recognition, Face Recognition, Speaker Recognition, Iris Recognition, Hand Geometry, Signature Verification. Introduction to Information Hiding: Technical Steganography, Linguistic Steganography, Copy Right Enforcement, Wisdom from Cryptography.

MODULE-II

Principles of Steganography: Framework for Secret Communication, Security of Steganography System, Information Hiding in Noisy Data, Adaptive versus non-Adaptive Algorithms, Active and Malicious Attackers, Information hiding in Written Text. A Survey of Steganographic Techniques: Substitution systems and Bit Plane Tools, Transform Domain Techniques: - Spread Spectrum and Information hiding, Statistical Steganography, Distortion Techniques, Cover Generation Techniques. Steganalysis: Looking for Signatures: - Extracting hidden Information, Disabling Hidden Information.

MODULE-III

Watermarking and Copyright Protection: Basic Watermarking, Watermarking Applications, Requirements and Algorithmic Design Issues, Evaluation and Benchmarking of Watermarking system. Computer Forensics, Rules of evidence, Evidence dynamics, Evidence collection, Data recovery, Preservation of digital evidence, surveillance tools for future warfare, cyber crime, types of cyber crimes, Digital evidence, nature of digital evidence, precautions while dealing with digital evidence.

Course Outcomes: After the completion of the course the student will be able to:

CO1	List the services performed by a crime investigators, crime laboratories and medical examiners.
CO2	Review the history and development of the forensic science sub-disciplines covered.
CO3	Discuss the role of a forensic scientist.
CO4	Familiarize oneself with the organization of a forensic science laboratory.
CO5	Students will able to recover data by using various techniques

Text Book:

1. Katzendbisser, Petitcolas, " Information Hiding Techniques for Steganography and Digital Watermarking", Artech House.